

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and**

trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = 150 MB = 150,000,000 bytes
- Segment size = 1500 bytes
- Data Link overhead = 40 bytes/frame
- Bandwidth = 50 Mbps

1. Number of segments

$$\frac{150,000,000}{1500} = 100,000$$

Answer = 100,000 segments (100,000 frames)

2. Total overhead

$$100,000 \times 40 = 4,000,000 \text{ bytes} = 4 \text{ MB}$$

Answer = 4 MB

3. Total transmission time

Total data transmitted

$$150 + 4 = 154 \text{ MB}$$

Convert to bits

$$154 \times 8 = 1232 \text{ Mb}$$

$$\frac{1232}{50} = 24.64 \text{ s}$$

Answer = 24.64 seconds

Transport Layer: Segmentation, flow control, error recovery.

Data Link Layer: Framing, error detection, reliable node-to-node delivery.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Total frames = 48
- Frame size = 1024 bytes

1. Number of transmission windows

$$\frac{48}{6} = 8$$

Answer = 8 windows

2. Total retransmissions

1 frame lost per window

$$8 \times 1 = 8$$

Answer = 8 retransmissions

Flow Control: Regulates the sender's transmission rate to prevent receiver buffer overflow, improving efficiency and reducing packet loss.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes/fragment

1. Number of fragments

Payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

$$\left\lceil \frac{12000}{1480} \right\rceil = 9$$

Answer = 9 fragments

2. Total header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Answer = 180 bytes

Network Layer: Fragments packets according to the MTU, routes them to the destination, and enables reassembly of the original packet at the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Total frames = 48
- Frame size = 1024 bytes

1. Number of transmission windows

$$\frac{48}{6} = 8$$

Answer = 8 windows

2. Total retransmissions

$$8 \times 1 = 8$$

Answer = 8 retransmissions

Flow Control: Controls the sender's transmission rate to match the receiver's capacity, preventing buffer overflow and improving communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- Video file = 600 MB
- Checkpoint after every 60 MB
- Failure after 390 MB

1. Number of checkpoints before failure

$$\frac{390}{60} = 6.5$$

Completed checkpoints = 6

Answer = 6 checkpoints

2. Data to be retransmitted

Last checkpoint =

$$6 \times 60 = 360 \text{ MB}$$

Retransmission

$$390 - 360 = 30 \text{ MB}$$

Answer = 30 MB

Synchronization: The Session Layer creates checkpoints during transmission so that after a failure, communication resumes from the last checkpoint instead of restarting the entire transfer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original file size = 900 MB
- Compression = 40%
- Encryption overhead = 5%

1. Compressed file size

$$900 - (40\% \times 900)$$
$$900 - 360 = 540 \text{ MB}$$

Answer = 540 MB

2. Encrypted file size

$$540 + (5\% \times 540)$$
$$540 + 27 = 567 \text{ MB}$$

Answer = 567 MB

3. Total percentage reduction

$$\frac{900 - 567}{900} \times 100$$
$$= \frac{333}{900} \times 100 = 37\%$$

Answer = 37% reduction

Presentation Layer: Compresses data to reduce transmission size and encrypts data to ensure confidentiality and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides

reliable services to users.

Given:

- Data = 3 GB = 3000 MB
- Throughput = 120 Mbps
- Request size = 3 MB

1. Number of requests

$$\frac{3000}{3} = 1000$$

Answer = 1000 requests

2. Transmission time

$$3000 \times 3 = 9000 \text{ MB}$$

$$\frac{9000}{120} = 75 \text{ s}$$

Answer = 75 seconds

Application Layer: Provides services such as file transfer (FTP), user authentication, error handling, and reliable communication between applications.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given:

- Application Layer = 4 ms
- Transport Layer = 3 ms
- Network Layer = 5 ms
- Data Link Layer = 2 ms
- Physical Layer = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

Total end-to-end delay

$$4 + 3 + 5 + 2 + 1 + 18 + 12 = 45 \text{ ms}$$

Answer = 45 ms

OSI Layer Contribution:

- Application Layer – Provides network services to applications.
- Transport Layer – Ensures reliable end-to-end communication.
- Network Layer – Performs routing and packet forwarding.
- Data Link Layer – Provides framing and error detection.
- Physical Layer – Transmits data as signals over the medium.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead = 80 bytes/frame

1. Number of frames

Useful data per frame

$$1600 - 80 = 1520 \text{ bytes}$$

$$\left\lceil \frac{80,000,000}{1520} \right\rceil = 52,632$$

Answer = 52,632 frames

2. Total overhead

$$52,632 \times 80 = 4,210,560 \text{ bytes}$$

Answer = 4,210,560 bytes (≈ 4.21 MB)

3. Transmission efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer = 95%

Protocol Overhead: Protocol headers and trailers increase the amount of transmitted data, reducing transmission efficiency and increasing network bandwidth usage.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to

ensure secure and reliable communication.

Given:

- Original file size = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Data Link overhead = 60 bytes/frame

1. Compressed file size

$$240 - (30\% \times 240)$$

$$240 - 72 = 168 \text{ MB}$$

Answer = 168 MB

2. Number of segments

$$168 \text{ MB} = 168,000,000 \text{ bytes}$$

$$\frac{168,000,000}{1400} = 120,000$$

Answer = 120,000 segments

3. Total protocol overhead

$$120,000 \times 60 = 7,200,000 \text{ bytes}$$

Answer = 7,200,000 bytes (7.2 MB)

4. Final amount of data transmitted

$$168 + 7.2 = 175.2 \text{ MB}$$

Answer = 175.2 MB

OSI Layers: The Presentation Layer compresses data, the Transport Layer performs segmentation, the Data Link Layer adds framing and error detection, and the Physical Layer transmits the data over the communication medium.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

and
coherence

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
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Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand
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